Musanna Galib

Postdoctoral Appointee at Argonne National Laboratory
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Website , LinkedIn , Github , Google Scholar , Orcid

EDUCATION

Ph.D. • Mechanical Engineering (CGPA: 87%) [Sep 2020 - Dec 2025]
4-Year Doctoral Fellow, UBC, Vancouver, BC, Canada
Dissertation: Dendrite Inhibition Strategy using Hetero-epitaxy in Thin Film Deposition Mechanics

M.Sc. • Mechanical Engineering (CGPA: 3.92/4.00) [Apr 2017 - May 2019]
BUET, Dhaka, Bangladesh
Dissertation: Study of electro-mechanical properties of manganese-based nanocomposites for super-capacitors

B.Sc. • Mechanical Engineering (CGPA: 3.92/4.00; Rank: 4th in 180) [Apr 2012 - Feb 2017]
BUET, Dhaka, Bangladesh
Dissertation: Computer Modeling of Fire Smoke Movement in Multizone Structure Using Two Open Source Platforms

WORK EXPERIENCE

Postdoctoral Appointee [Jan 2026 - Present]
Energy Storage Research Alliance
Argonne National Laboratory, USA

Graduate Research Assistant [Oct 2021 - Dec 2025]
Dept. of Mechanical Engineering, UBC, Canada

Graduate Teaching Assistant (6 courses)
Dept. of Mechanical Engineering, UBC, Canada [Sep 2022 - Apr 2025]
Dept. of Physics, UBC, Canada [Jan 2024 - Apr 2024]

Assistant Professor of Teaching (10 Courses) [Dec 2019 - Sep 2021]
Dept. of Mechanical Engineering, BUET, Bangladesh

Lecturer (13 Courses) [May 2017 - Dec 2019]
Dept. of Mechanical Engineering, BUET, Bangladesh

RESEARCH EXPERIENCE

Scientific Data Infrastructure Development [2026 - Present]
• ESRA FAIR data platform: ingestion pipeline, metadata schema, and scientific workflows, deployed at data.energystoragera.org

Software Development [2023 - 2025]
• Python package development at UBCV

Machine Learning Research [2023 - 2025]
• Interatomic potential development at Modeling and Simulation Research Group, UBCV

Battery Research [2020 - 2025]
• Battery recycling and Multi-scale modeling at Modeling and Simulation Research Group, UBCV
• Battery testing at Advanced Materials for Energy Storage Lab, UBCO

TECHNICAL SKILLS

Programming Languages: Python, MATLAB, Bash, C, HTML, CSS.
Deep Learning Packages: PyTorch.
Material Simulation Tools: VASP, LAMMPS.
FEM Modeling Tools: ABAQUS, MOOSE.
Visualization & Design Tool: Vesta, OriginPro, OVITO, Paraview, Atomsk, Solidworks.
Battery Testing: Battery cell assembly, Cyclic testing, In-situ optical microscopy.
Clean-room Testing: DEKTAK profilometer.
High-performance Computing (HPC): Alliance, ARC Server.

Documentation & Presentation Application: Office Suite, LaTeX.
Operating System: Windows (WSL), Linux (Ubuntu).

HONORS & AWARDS

- Institute of Applied Mathematics (IAM) travel grant [May 2025]
- 1st prize in "Ideas With Impact" Competition by The Institute for Computing, Information and Cognitive Systems (ICICS) (Link) [May 2024]
- CSME Congress "Best Presentation Award" in Computational Mechanics at University of Toronto [May 2024]
- Conference Travel Support for ICICS Graduate Students for SES-2023 [Aug 2023]
- Collaborative Research Mobility Award (UBC CRMA) in UBC [Jan - Dec 2023]
- 4-Year Doctoral Fellowship in UBC [Sep 2020 - Aug 2024]
- Faculty of Applied Science Graduate Award in UBC [Sep 2021]
- President's Academic Excellence Initiative Ph.D. Award in UBC [Sep 2020 - Apr 2025]
- International Tuition Award in UBC [Sep 2020 - Apr 2023]
- Bangladesh-Sweden Trust Fund (BSTF) Travel Grant for Ph.D. [2021]
- Dean's List for 4 consecutive years in BUET [Feb 2013 - Feb 2017]
- University Merit for 4 consecutive years in BUET [Feb 2013 - Feb 2017]
- Board Scholarship in HSC 2011 [Jan 2011]
- Board Scholarship in SSC 2009 (37^{th} (among 900k students) in Bangladesh) [Jan 2009]

GRANTS & FUNDING

- DRI EDIA Champions in Digital Research Alliance of Canada [Amount: 35,000 CAD]
  3.15 million funding opportunity (90 Grants) for graduate or post-graduate researchers in Canada (Link)
- 4-Year Doctoral Fellowship in UBC (Link) [Amount: 93,267 CAD]

MEDIA COVERAGE

- Media coverage from the University of British Columbia for Ph.D. research (Link) [2025]
- Journal Cover in ACS Applied Materials & Interfaces (Link) [2025]
- Announcement for 1st prize in "Ideas With Impact" Competition by UBC (Link) [2024]

PUBLICATION

Journal Publication

- **Galib, M.**, Isiet, M., and Ponga, M., 'AtomProNet: Data flow to and from machine learning interatomic potentials in materials science', Machine Learning: Science and Technology, 7, 3, 035056, 2026. (Link)
- **Galib, M.**, Xu, J., Amardeep, A., Liu, J., and Ponga, M., 'Dendrite Suppression in Zn Batteries Through Hetero-Epitaxial Residual Stresses Shield' , ACS Applied Materials & Interfaces, 17, 35, 49387–49399, 2025. (Link)
- Isiet, M., **Galib, M.**, Xiao, Y., Dadap, J. I., Ye, Z., and Ponga, M., 'Spall failure of alumina at high-strain rates using femtosecond laser experiments and high-fidelity molecular dynamics simulations', Acta Materialia, 299, 121436, 2025. (Link)
- **Galib, M.**, Orhan, K.O., Liu, J., and Ponga, M.,'Residual Stress Development in Lattice Mismatched Epitaxial Thin Films via Atomic and Molecular Layer Depositions', Journal of the Mechanics and Physics of Solids, 193, 105897, 2024. (Link)
- Xu, J.*, **Galib, M.***, Wu, Z., Tao, L., Shao, Y., Zhang, Y., Guo, X., Hansen, E. J., Chen, Y., Wang, Z., Liu, C., Ponga, M., and Liu, J.,' High-entropy Strategy to Suppress Volumetric Strain and Enhance Diffusion Rate of $Na_3V_2(PO_4)_2F_3$ Cathode for Durable and High-areal-capacity Zinc-ion Battery Pouch Cells', Nano Energy, 110373, 2024. (*Co-first authors) (Link)

- Khan, A.A., Rabi, S.S., Jamee, T. **Galib, M.**, Ashek, F.E., and Rahman, M.A., 'Effects of Crumpling Stage and Porosity of Graphene Electrode on the Performance of Electrochemical Supercapacitor,' J. Phys. Chem. B, 128, 39, 9586–9597, 2024. (Link)
- **Galib, M.**, Orhan, K.O., and Ponga, M., 'Engineering Chemo-Mechanical Properties of Zn Surfaces via Alucone Coating', J. Phys. Chem. C, 127, 5, 2481–2492, 2023. (Link)
- **Galib, M.**, Hosen, M.M., Saha, J.K., Islam, M.M., Firoz, S.H., and Rahman, M.A.,'Electrode Surface Modification of Graphene-MnO_2 Supercapacitors using Molecular Dynamics Simulations', Journal of Molecular Modeling, 26, 251, 2020. (Link)
- Saha, P., Ishtiaque, M.M.U., Sutradhar, A., **Galib, M.**, and Hannan, M.A., 'A sustainable approach to improve the efficiency of Earth Pipe Cooling System', International Journal of Sustainable Engineering, 13, 5, 387-397, March 2020. (Link)

Conference Publication

- Arka, A.M., Mridha, R.H., Shafqat, R., **Galib, M.**, and Morshed, AKM M., 'Design and Comparative Parametric Analysis Using NSGA-II for Multivariable Constrained Optimization of Shell and Tube Heat Exchangers', 13th International Conference on Mechanical Engineering (ICME), December 2019, Dhaka, Bangladesh (Link)
- Saha, P., **Galib, M.**, Ishtiaque, M.M.U., Akanda, S.R., and Hannan, M.A., 'Numerical Study on Improving the Efficiency of the Earth Pipe Cooling System', International Conference on Civil and Environmental Engineering , November 2018, Kuala Lumpur, Malaysia. (Link)
- Khandoker, M.A.R., **Galib, M.**, Islam, A., and Rahman, M.A., 'Modeling of smoke movement in multizone garments building using two open source platforms', The 7^{th} BSME International Conference on Thermal Engineering 2016, June 2017, Dhaka, Bangladesh. (Link)
- Chatterjee, A., **Galib, M.**, and Sarkar, M.A.R., 'Application of Arduino in designing modern electromechanical laboratory', 2017 IEEE International Conference on Power, Control, Signals and Instrumentation Engineering (ICPCSI), June 2018, Chennai, India. (Link)

Poster Presentation

- **Galib, M.**, and Chan, M.K., 'ESRA: A Community Effort to Build a FAIR Data Framework for Energy Materials,' U.S.-Japan Workshop: AI for Energy Storage Materials, 27 August 2026, Lemont, Illinois, USA.
- **Galib, M.**, Liu, J., and Ponga, M., 'Dendrite Inhibition Strategy using Hetero-epitaxial Residual Stresses in Thin Film Deposition Mechanics,' 247th ECS Meeting, 19 May 2025, Montreal, Québec.
- **Galib, M.**, Liu, J., and Ponga, M., 'Dendrite Inhibition Strategy using Hetero-epitaxial Residual Stresses in Thin Film Deposition Mechanics', 16th World Congress on Computational Mechanics (WCCM), 24 July 2024, Vancouver, British Columbia.

Book Chapter

- Pelletier, J., Wolf, J., Scheer, E.R., Arora, J., **Galib, M.**, and Valdez, R., 'Engineering Solutions to Equity, Diversity, and Inclusion' chapter in the book 'Equity, Diversity & Inclusion: Sisyphean Undertaking to an Achievable Reality', 2026 (Link).

SOFTWARE & PLATFORMS

Research Platforms

- ESRA Data Catalog: FAIR data platform for energy storage materials at Argonne (Link)

Python Packages

- AtomProNet: Atomic Data Processing for Neural Network (Link)
- SurfTrack: Dendrite tracking from in-situ optical microscopy (Link)
- MOOSEanalyze: Post-processing software for exodus file (Link)

CONTRIBUTED TALKS

- 'Dendrite Inhibition Strategy using Hetero-epitaxial Residual Stresses in Thin Film Deposition Mechanics' [25 July 2024]
  16th World Congress on Computational Mechanics, Vancouver, BC, 2024
- 'Multiscale Approach to Study High Strain Rate Deformation' [27 May 2024]
  Canadian Society for Mechanical Engineering, Toronto, Ontario, 2024
- 'Dendrite Inhibition Strategy using Hetero-epitaxial Residual Stresses in Thin Film Deposition Mechanics' [9 October 2023]
  Society of Engineering Science Annual Technical Meeting, Minneapolis, Minnesota, 2023
- 'Using Thin Film to Improve Metal Anode Battery Safety' [9 March 2023]
  Top presenter, Open heat - 3 Minutes Thesis (3MT), Vancouver, UBC, 2023
- 'Residual Stresses in Thin Film Deposition Mechanics' [19 October 2022]
  Society of Engineering Science Annual Technical Meeting, College Station, Texas, 2022

PROFESSIONAL EXPERIENCES

- NASA Open Science 101, NASA Science Mission Directorate (Link) [Mar 2025]
- CIRTL Associate, Center for the Integration of Research, Teaching and Learning [Oct 2024]
- Member, Battery Division, The Electrochemical Society (ECS) [2024-2025]
- Member, Canadian Society for Mechanical Engineering (CSME) [2024-2025]
- Member, Board of Undergraduate Studies of Mechanical Engineering, BUET [2017-2021]
- Member, Bureau of Research, Testing and Consultation (BRTC), BUET [2019-2021]
- Member, Publication and Secretariat committee, The 12^{th} & 13^{th} International Conference on Mechanical Engineering (ICME 2017 (Link) & 2019 (Link)) [2017, 2019]

REVIEWER

- Additive Manufacturing; Journal of Energy Storage; Journal of Applied Physics

WORKSHOPS & TRAINING ATTENDED

- Instructional Skills Workshop by Centre for Teaching, Learning, and Technology at UBC [Oct 2024]
- Project Management by Mitacs Inc., Canada [Nov 2023]

EXTRA-CURRICULAR ACTIVITIES

- Steward, UBC Teaching Assistants' & Instructors Union (CUPE 2278) [2024-2025]
- Vice President (External), Bangladeshi Grad Alliance (BGA) at UBC [2024-2025]
- Vice President (Academic), Mechanical Engineering Graduate Association, UBC [2022-2023]
- Award Leader, The Duke of Edinburgh's Award Foundation Bangladesh [2017-2021]
- Gold Awardee of 'The Duke of Edinburgh's Award [2017]
- Financial Secretary at BUET Self Defense Club [2016-17]
- Member at United Nations Youth and Students Association of Bangladesh [2013-16]

REFERENCES

Available upon request.